

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended.

16 Regulatory Status

GRAS listed. Accepted for use in human and veterinary parenteral vaccines in Europe and the USA. The limits for use in human vaccines are 0.85 mg aluminum/dose (FDA) and 1.25 mg aluminum/dose (WHO). There are no established limits for use in veterinary vaccines. Reported in the EPA TSCA Inventory.

17 Related Substances

Aluminum phosphate adjuvant.

18 Comments

Different grades of aluminum hydroxide adjuvant with various concentrations, protein binding capacities, and points of zero charge are available.

The impurity limits at 2% equivalent Al_2O_3 are $\text{Cl} < 0.5\%$; $\text{SO}_4 < 0.5\%$; $\text{PO}_4 < 0.1\%$; $\text{NO}_3 < 0.1\%$; $\text{NH}_4 < 0.1\%$; $\text{Fe} < 20 \text{ ppm}$; $\text{As} < 0.6 \text{ ppm}$; and heavy metals $< 20 \text{ ppm}$.

The aluminum hydroxide gel referred to in the USP 32 is used in cosmetics as an emollient, filler, humectant, a mild astringent, and viscosity controlling agent. In pharmaceutical preparations it is used as an adsorbent, and as a protein binder.⁽⁵⁾ It is also used therapeutically as an antacid, and as an abrasive in dentifrices. It is not, however, used as a vaccine adjuvant.

19 Specific References

- 1 Shirodkar S *et al.* Aluminum compounds used as adjuvants in vaccines. *Pharm Res* 1990; 7: 1282–1288.

- 2 Prowse CV *et al.* Changes in factor VIII complex activities during the production of a clinical intermediate purity factor VIII concentrate. *Thromb Haemost* 1981; 46: 597–601.
- 3 Johnston CT *et al.* Measuring the surface area of aluminum hydroxide adjuvant. *J Pharm Sci* 2002; 91: 1702–1706.
- 4 Goldenthal KL *et al.* Safety evaluation of vaccine adjuvants. *AIDS Res Hum Retroviruses* 1993; 9(Suppl. 1): S47–S51.
- 5 Ash M, Ash I. *Handbook of Pharmaceutical Additives*, 3rd edn. Endicott, NY: Synapse Information Resources, 2007; 446.

20 General References

- Gupta RK *et al.* Adjuvant properties of aluminum and calcium compounds. Powell MF, Newman MJ, eds. *Vaccine Design*. New York: Plenum, 1995; 229–248.
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- Vogel FR, Hem SL. Immunogenic adjuvants. Plotkin SA *et al.*, ed. *Vaccines*, 5th edn. New York: W.B. Saunders, 2008; 59–71.
- White JL, Hem SL. Characterization of aluminum-containing adjuvants. Brown F *et al.*, ed. *Physico-Chemical Procedures for the Characterization of Vaccines*. IABS Symposia Series, Development in Biologicals, vol. 103. New York: Karger, 2000; 217–228.

21 Authors

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22 Date of Revision

19 February 2009.

Aluminum Monostearate

1 Nonproprietary Names

JP: Aluminum Monostearate

USP-NF: Aluminum Monostearate

2 Synonyms

Aluminum stearate; aluminum, dihydroxy (octadecanoate-O-); dihydroxyaluminum monostearate; *HyQual*; octadecanoic acid aluminum salt; stearic acid aluminum salt; stearic acid aluminum dihydroxide salt; *Synpro*.

3 Chemical Name and CAS Registry Number

Aluminum monostearate [7047-84-9]

4 Empirical Formula and Molecular Weight

$\text{C}_{18}\text{H}_{37}\text{AlO}_4$ 344.50

5 Structural Formula

$[\text{CH}_3(\text{CH}_2)_{16}\text{COO}]\text{Al}(\text{OH})_2$

6 Functional Category

Emollient; emulsion stabilizer; gelling agent; opacifier; stabilizing agent.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum monostearate is mainly used in microencapsulation⁽¹⁻³⁾ and in the manufacture of ointments. Aluminum monostearate is used as a viscosity-increasing agent in nonaqueous cosmetic and pharmaceutical formulations. In addition, aluminum monostearate can be used as an emulsion stabilizer in cosmetic emulsions and is used in cosmetics such as mascara, moisturizers, and sunscreens.

8 Description

Aluminum monostearate is an aluminum compound of stearic acid and palmitic acid. The USP32–NF27 states that aluminum monostearate contains the equivalent of not less than 14.5% and not more than 16.5% of Al_2O_3 , calculated on the dried basis. The JP XV states that it contains not less than 7.2% and not more than 8.9% of aluminium.

Aluminum monostearate occurs as a white, fine, bulky powder with a slight odor of fatty acid. It is a solid material.

9 Pharmacopeial Specifications

See Table I. See also Section 18.

Table I: Pharmacopeial specifications for aluminum monostearate.

Test	JP XV	USP32-NF27
Identification	+	+
Description	≤3.0%	≤2.0%
Loss on drying	+	—
Arsenic	≤2 ppm	≤4 ppm
Heavy metals	≤50 ppm	≤50 ppm
Acid value for fatty acid	+	—
Free fatty acid	+	—
Water-soluble salts	≤10 mg	—
Assay of Al (dried basis)	7.2–8.9%	14.5–16.5%

10 Typical Properties

See Table II.

Melting point 220–225°C

Solubility Practically insoluble in water. Soluble in ethanol (95%) and benzene.

Specific gravity 1.14

Table II: Typical physical properties of selected commercially available aluminum monostearates.

Grade	Assay (as Al ₂ O ₃) (%)	Loss on drying (%)	Median particle size (μm)
Synpro Aluminum Monostearate NF	15.5	0.8	7.0
Synpro Aluminum Monostearate NF Gellant	15.3	1.6	—
HyQual Aluminum Monostearate NF Powder	14.5–16.5	≤2.0	—
HyQual Aluminum Monostearate NF Fine Powder	14.5–16.5	≤2.0	—

11 Stability and Storage Conditions

Aluminum monostearate should be stored in a well-closed container in a cool, dry, place. It is stable under ordinary conditions of use and storage.

12 Incompatibilities

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13 Method of Manufacture

Aluminum monostearate is prepared by reacting aluminum with stearic acid.

14 Safety

Aluminum monostearate is generally regarded as relatively non-toxic and nonirritant when used as an excipient.

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled. When heated to decomposition, aluminum monostearate emits acrid smoke and irritating vapors.

16 Regulatory Status

Aluminum monostearate and aluminum stearate are included in the FDA Inactive Ingredients Database (oral capsules and tablets, topical creams and ointments). Included in nonparenteral medicines licensed in the UK.

17 Related Substances

Aluminum distearate; aluminum tristearate.

Aluminum distearate

Empirical formula C₃₆H₃₇AlO₅

Molecular weight 877.39

CAS number [300-92-5]

Synonyms Hydroxyaluminum distearate; aluminum stearate; aluminum monobasic stearate.

Description Aluminum distearate occurs as a fine white to off-white colored powder with a slight odor of fatty acid.

Melting point 150–165°C

Specific gravity 1.01

Solubility Soluble in benzene, and in ethanol (95%); practically insoluble in water.

Comments The EINECS number for aluminum distearate is 206-101-8.

Aluminum tristearate

Empirical formula C₅₄H₁₀₅AlO₆

Molecular weight 610.9

CAS number [637-12-7]

Synonyms Hydroxyaluminum tristearate; aluminum stearate.

Description Aluminum tristearate occurs as a fine white to off-white colored powder with a slight odor of fatty acid.

Melting point 117–120°C

Specific gravity 1.01

Solubility Practically insoluble in water. Soluble in ethanol (95%), benzene, turpentine oil, and mineral oils when freshly prepared.

Comments The EINECS number for aluminum tristearate is 211-279-5.

18 Comments

A specification for aluminum stearate, described as consisting mainly of the distearate, is included in the *Japanese Pharmaceutical Excipients* (JPE).⁽⁴⁾

It should be noted that aluminum stearate can also refer to the distearate (CAS number 300-92-5) and the monostearate (CAS number 7047-84-9) in addition to the tristearate (CAS number 637-12-7). The distearate exhibits the same excipient properties as the tristearate and is used in similar pharmaceutical applications. However, the monostearate is more widely used in both cosmetic and pharmaceutical preparations.

The EINECS number for aluminum monostearate is 230-325-5.

19 Specific References

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- Wu PC *et al.* Preparation and evaluation of sustained release microspheres of potassium chloride prepared with ethylcellulose. *Int J Pharm* 2003; 260: 115–121.
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- Japan Pharmaceutical Excipients Council. *Japanese Pharmaceutical Excipients* 2004. Tokyo: Yakuji Nippo, 2004; 74–75.

20 General References

Ferro Corporation. Technical literature: *Synpro Aluminum Monostearate NF*, 2008.
 Mallinckrodt. Technical literature: *HyQual Aluminum stearate*, 2008.

21 Author

J Shur.

22 Date of Revision

19 February 2009.



Aluminum Oxide

1 Nonproprietary Names

None adopted.

2 Synonyms

Activated alumina; activated aluminum oxide; alpha aluminum oxide; alumina; alumina, calcined; alumina, tabular; aluminum oxide alumite; aluminum trioxide; gamma aluminum oxide.

3 Chemical Name and CAS Registry Number

Aluminum oxide [1344-28-1]

4 Empirical Formula and Molecular Weight

Al_2O_3 101.96

5 Structural Formula

Aluminum oxide occurs naturally as the minerals bauxite, bayerite, boehmite, corundum, diaspore, and gibbsite.

6 Functional Category

Adsorbent; dispersing agent.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum oxide is used mainly in tablet formulations.⁽¹⁾ It is used for decoloring powders and is particularly widely used in antibiotic formulations. It is also used in suppositories, pessaries, and urethral inserts. Hydrated aluminum oxide (see Section 18) is used in mordant dyeing to make lake pigments, in cosmetics, and therapeutically as an antacid.

8 Description

Aluminum oxide occurs as a white crystalline powder. Aluminum oxide occurs as two crystalline forms: α -aluminum oxide is composed of colorless hexagonal crystals, and γ -aluminum oxide is composed of minute colorless cubic crystals that are transformed to the α -form at high temperatures.

9 Pharmacopeial Specifications

See Section 18.

10 Typical Properties

Boiling point 2977°C
Density (bulk) 0.9–1.1 g/cm³
Flammability Nonflammable.
Hardness (Mohs) 8.8
Hygroscopicity Very hygroscopic.

Melting point 2050°C

Solubility Slowly soluble in aqueous alkaline solutions with the formation of hydroxides; practically insoluble in nonpolar organic solvents, diethyl ether, ethanol (95%), and water.

Specific gravity 2.8 (becomes 4.0 at 800°C)

Vapor pressure 133.3 Pa at 2158°C

11 Stability and Storage Conditions

Aluminum oxide should be stored in a well-closed container in a cool, dry, place. It is very hygroscopic.

12 Incompatibilities

Aluminum oxide should be kept well away from water. It is incompatible with strong oxidizers and chlorinated rubber. Aluminum oxide also reacts with chlorine trifluoride, ethylene oxide, sodium nitrate, and vinyl acetate. Exothermic reactions above 200°C with halocarbon vapors produce toxic hydrogen chloride and phosgene fumes.

13 Method of Manufacture

Most of the aluminum oxide produced commercially is obtained by the calcination of aluminum hydroxide.

14 Safety

Aluminum oxide is generally regarded as relatively nontoxic and nonirritant when used as an excipient. Inhalation of finely divided particles may cause lung damage (Shaver's disease).⁽²⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled.⁽³⁾ In the UK, the workplace exposure limits for aluminum oxide are 10 mg/m³ long-term (8-hour TWA) for total inhalable dust and 4 mg/m³ for respirable dust.⁽⁴⁾ In the USA, the OSHA limit is 15 mg/m³ total dust, 5 mg/m³ respirable fraction for aluminium oxide.⁽⁵⁾

16 Regulatory Status

Included in the FDA Inactive Ingredients Database (oral tablets and topical sponge). Included in nonparenteral medicines licensed in the UK.

17 Related Substances

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18 Comments

A specification for aluminum oxide is included in the *Japanese Pharmaceutical Excipients* (JPE);⁽⁶⁾ see Table I. A specification for